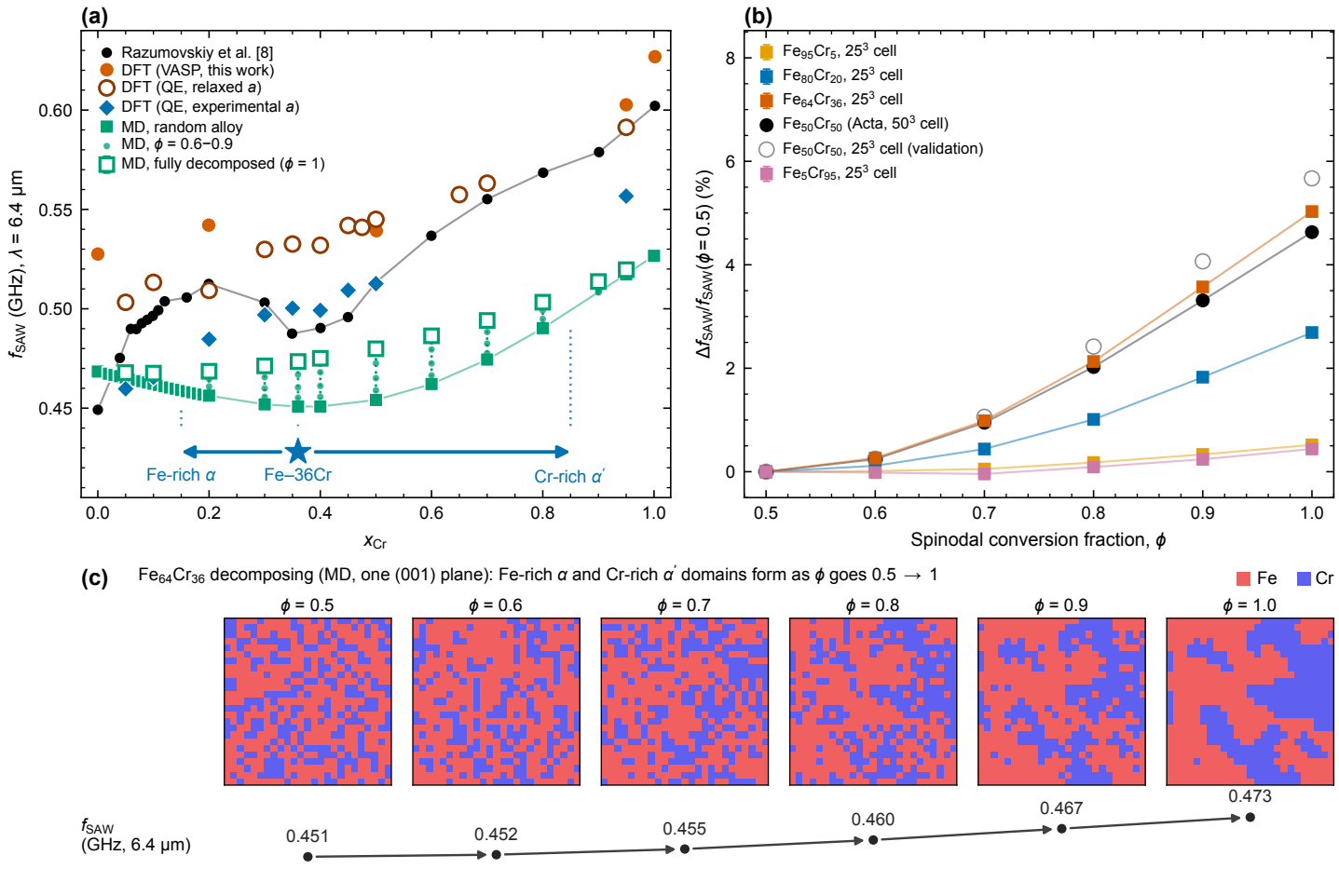

(a)
Razumovskiy et al. [8]
DFT (VASP, this work)
DFT (QE, relaxed a)
DFT (QE, experimental a)
MD, random alloy
MD, ϕ = 0.6–0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α′
$f_{SAW}$ (GHz), λ = 6.4 μm
$x_{Cr}$
(b)
$Fe_{95}Cr_5$, $25^3$ cell
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$\Delta f_{SAW}/f_{SAW}(\phi = 0.5)$ (%)
Spinodal conversion fraction, ϕ
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, one (001) plane): Fe-rich α and Cr-rich α′ domains form as ϕ goes 0.5 → 1
Fe
Cr
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
$f_{SAW}$ (GHz, 6.4 μm)
0.451
0.452
0.455
0.460
0.467
0.473